

Medical Audio vs Transcript Evaluation Framework

Technical Report: Problem, Design Evolution, Final Architecture, and Evaluation

1. Problem and Motivation

Medical Speech-to-Text (STT) cannot be evaluated reliably using Word Error Rate (WER) alone, because a small transcription error can change clinical meaning. **20 mg** may become **200 mg**, or **requires dialysis** may become **does not require dialysis**. WER treats such errors the same as harmless spelling noise, even though one serious error can matter more than several minor ones.

This project sets out to build a **medical transcript evaluation system** (not a new STT model) that checks whether a transcript correctly represents spoken content, targeting four error types:

- **Missing:** spoken information absent from the transcript.
- **Incorrect:** a value (dose, date, name) is wrong.
- **Conflicting:** meaning is changed, e.g. a negation.
- **Hallucinated:** content not spoken appears in the transcript.

2. Initial Approach: Gold-Reference Audit

The first idea was straightforward: compare an STT transcript against a trusted human transcript. Audio is transcribed locally with **faster-whisper**, and both transcripts are normalized, segmented, and aligned. From this alignment the system computes WER, CER, token coverage, hallucination ratio, and checks on numbers, dates, doses, and times.

This worked, but it revealed a limitation: WER and CER are useful signals, but they do not capture **clinical severity**. A transcript could score well overall while still containing one dangerous local error. To address this, the scoring was changed to weight medical entity errors and severity more heavily than generic text differences, so a good overall score cannot hide a serious mistake. Unclear differences are classified by DeepSeek; a transcript is marked **Mismatch** when the final score falls below 90 or a high-severity error is detected.

3. Challenge: What If There Is No Gold Transcript?

Creating a trusted human transcript for every audio file is expensive and slow, which limits how widely Approach 1 can be applied. This led to a second approach: transcribe the same audio independently with **Whisper and Deepgram**, and compare their outputs to each other instead of to a human reference.

A second problem appeared immediately: the two engines produce **different segment boundaries**, so segment 1 from one engine does not necessarily correspond to segment 1 from the other. Simple segment-to-segment comparison was not reliable. This was solved with a pipeline of **word-level normalization** → **global alignment** (a Needleman-Wunsch-style method with a time penalty, so words spoken close together in time are more likely to be matched) → **disagreement detection**, identifying insertions, deletions, substitutions, and matches between the two transcripts.

4. Challenge: Agreement Does Not Mean Correctness

Two-engine agreement is a strong signal, but it is not a guarantee. If both Whisper and Deepgram output *"patient requires dialysis"*, they agree, but both could still be wrong, particularly on unfamiliar medical terms that both engines are likely to mis-hear in the same way. **Two-engine agreement provides evidence, not ground truth**. This insight shaped the final design: agreement, average word confidence, and critical-error rules (negation changes, number changes, medical term changes, missing or added important words, large technical substitutions) are combined into a single **risk score**. Segments with high risk are escalated for additional verification rather than accepted automatically.

5. Final Architecture

Audio → STT → Normalize/Align → Deterministic Checks → Risk Score → Unclear or high-risk?
→ **No:** Auto-accept / Review → **Yes:** Multimodal LLM → Human Review

The guiding principle across both approaches is the same: use simple, explainable checks first, and escalate to an LLM or a human only when the case is unclear or the error may be serious. Two concrete evaluation modes result from this design:

1. **Gold Reference:** compares an STT transcript with a trusted human transcript (Section 2).
2. **Two-Engine Consensus:** compares independent Whisper and Deepgram transcripts when no human transcript exists (Sections 3-4).

Aspect	Approach 1: Gold Reference	Approach 2: Two-Engine Consensus
Input	Audio + trusted human transcript	Audio only
Comparison	STT vs. human transcript	Whisper vs. Deepgram
LLM role	DeepSeek on unclear differences	Gemini on flagged segments
Strength	Strong verification when reference is reliable	Works without a human transcript
Weakness	Reference transcripts are costly to build	Both engines may share the same error

6. Technical Implementation

6.1. Approach 1: Gold-Reference Scoring Pipeline

The Gold-Reference pipeline runs as a fixed sequence: transcribe the audio with **faster-whisper**; normalize both the generated and human transcripts; segment and align the corresponding text; compute WER, CER, token coverage, hallucination ratio, and medical entity checks (numbers, dates, doses, times); classify unclear differences with DeepSeek or rule-based checks; then combine all of this into a single final score. The combination step is weighted so that medical entity and severity errors count for more than generic text differences, which is what allows a transcript with a good overall score to still be flagged when it contains one dangerous local error. As described in Section 2, a transcript is marked **Mismatch** when the final score falls below 90 or a high-severity error is detected.

6.2. Approach 2: Two-Engine Confidence Score

For Approach 2, the confidence score combines three signals:

$$C = 100 \times (0.40M + 0.45A + 0.15(1 - L))$$

where M = average word confidence, A = agreement between engines, and L = ratio of low-confidence words. This score maps to three action tiers:

Confidence	Tier	Action
≥ 98	<code>auto_accept</code>	Accept automatically
90-97	<code>review_technical</code>	Review technical terms
< 90	<code>mandatory</code>	Human verification required

7. Multimodal Verification and Human Review

The remaining question was how to catch errors that both STT engines make in the same way. Rather than claiming one model is simply "better," the system treats this as an additional verification layer: for selected high-risk segments, the **raw audio itself**, not just the text, is sent to a multimodal model (Gemini) for an independent interpretation. Gemini does not replace the STT engines; it is only consulted when deterministic checks and engine agreement leave the result unclear or high-risk. The final safeguard is a human review step, where flagged segments are presented with audio playback and a correction interface.

8. Evaluation and Results

Real demonstration data was kept separate from a controlled synthetic test set covering eight scenarios: perfect matches, number changes, negation changes, missing content, extra content, technical word corruption, contraction equivalence, and harmless spelling noise.

Method	Result	Interpretation
Approach 1: heuristic only	5/8 $\approx$ 62%	Detects numbers/negations without an LLM, but misses some missing, extra, and corrupted cases.
Approach 1: full pipeline	8/8 = 100%	With correct judge decisions, all cases map to the correct error category.
Approach 2: disagreement detector	8/8 = 100%	Critical-error rules correctly detect and label all test cases.

These results confirm the evaluation logic works on the selected synthetic cases; they **do not represent real hospital-scale medical accuracy**. A larger human-labeled dataset would be needed for that. Automated tests cover alignment, normalization, scoring, comparison, pipeline behavior, judge logic, review functions, API behavior, and end-to-end scenarios; live LLM tests are kept separate so the main suite runs without external API keys.

9. Challenges, Limitations, and Future Work

Remaining **limitations**: the real dataset is small; medical entity detection is still basic; parts of the system depend on external APIs; LLM usage adds cost and latency; LLMs can still make incorrect decisions; and engine agreement does not guarantee correctness.

Challenge	What was learned / solution
WER treats all errors similarly	Added severity-aware checks
No gold transcript available	Added two-engine consensus
Different STT segment boundaries	Used global word alignment
Both STTs can make the same mistake	Added audio-level multimodal verification
LLM calls are expensive	Send only suspicious segments
High average score can hide serious errors	Added per-segment risk tiers
Synthetic tests can look perfect	Clearly separated test validation from clinical accuracy

Future work includes building a larger human-labeled medical audio dataset, adding a drug/procedure glossary, measuring cost and latency per hour of audio, testing other languages and accents, and adding human or gold-reference verification for high-risk Approach 2 cases.

10. Conclusion

This project began with a simple observation: WER is not enough for medical transcripts. That observation led first to a severity-aware, gold-reference audit, then, once the cost of building reference transcripts became clear, to a two-engine consensus approach with global alignment. Recognizing that agreement between engines is evidence rather than proof led to a final design where deterministic checks, engine agreement, and a risk score decide when to escalate to multimodal verification and human review. The result is useful for demonstration, testing, and research; it is **not a certified medical device** and should not replace clinician review of important clinical notes.